AUTONOMOUS NETWORK FOR REAL-TIME WATER FLOW VELOCITY MONITORING USING LORAWAN

AUTHORS

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RELATED LITERATURE

1. Djalilov, A., Nazarov, O., Sobirov, E., & Tasheva, U. (2023). Research of water flow measuring device based on Arduino platform
2. Pearce, R. H., Chadwick, M. A., Main, B., Chan, K., Sayer, C. D., & Patmore, I. R. (2024). Low-cost approach to an instream water depth sensor construction using differential pressure sensors and Arduino microcontrollers

Water management requires continuous and spatially distributed data to prevent flooding, support ecosystem restoration, and optimize resource use. Traditional monitoring stations are often prohibitively expensive to deploy or require frequent maintenance, limiting their use in remote or restored environments. In peatland rewetting areas, monitoring typically focuses on water levels, while flow dynamics remain insufficiently captured. This project proposes a low-cost, energy-efficient sensor network for measuring water flow velocity in channels. By combining edge processing, LPWAN communication, and scalable data integration, the system provides near real-time hydraulic data suitable for environmental monitoring and geospatial analysis

INTRODUCTION

Peatland ecosystems play a crucial role in carbon storage and climate regulation. However, many peatland areas in Europe have been historically drained for forestry and agricultural use, leading to increased carbon emissions and disrupted hydrological systems.
Recent initiatives, such as peatland rewetting projects in regions like Luntener Wald, aim to restore natural water dynamics and improve carbon retention. These projects require continuous monitoring of hydrological parameters, including water levels and flow dynamics.
Currently, monitoring systems often focus on static measurements, such as groundwater level or water surface height at specific points. However, water flow velocity remains an under-monitored parameter, despite its importance for understanding water balance and restoration effectiveness.

HARDWARE

The physical sensing layer is designed for maximum autonomy in remote outdoor environments, operating on a solar-assisted battery system.

Adafruit ADA5066 Sensor: A G 1/2" threaded in-line turbine equipped with a magnet. A Hall effect sensor detects rotations through a waterproof plastic barrier.

Flow Metrics: Operates within a flow range of 1-30 Liters/min. Each electrical pulse output exactly corresponds to 2.25 ml of liquid passing through.

ESP32-S3 Microcontroller: Employs an Ultra-Low Power (ULP) strategy. The power-hungry main CPU sleeps while the coprocessor continuously counts hardware interrupts from the sensor.

METHODOLOGY

The system measures water flow velocity using a sensor installed directly in a channel. The sensor generates pulse signals proportional to the flow rate, which are processed by a microcontroller.

- To comply with LoRaWAN constraints, measurements are aggregated over fixed time intervals (e.g., 15 minutes). The total volume is calculated as the number of pulses multiplied by 2.25 milliliters per pulse.
- Encoded data is transmitted via LoRaWAN using The Things Network infrastructure. The payload is formatted using the CayenneLPP standard for efficient data representation.
- Timestamps are assigned on the network side upon packet reception, eliminating the need for onboard time synchronization and reducing device energy consumption.
- The data is then forwarded via an HTTP webhook and integrated into openSenseMap for visualization and sharing.

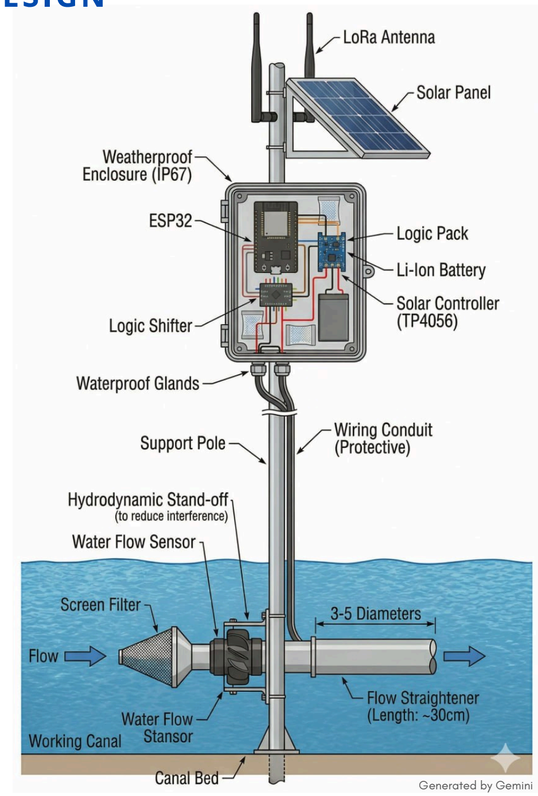
RESULTS

A preliminary test confirmed basic system functionality, including signal recording, data conversion, transmission via The Things Network, and integration into openSenseMap.
Due to technical issues, only limited data was collected, preventing full validation. However, the results demonstrate the feasibility of the approach.

PERFORMANCE

The system was tested under controlled and semi-natural conditions to evaluate:

- measurement stability
- signal transmission reliability
- consistency of recorded values

Preliminary results indicate that the sensor provides stable and continuous measurements of flow velocity. Data transmission via LoRaWAN proved to be reliable over long distances with low energy consumption.

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